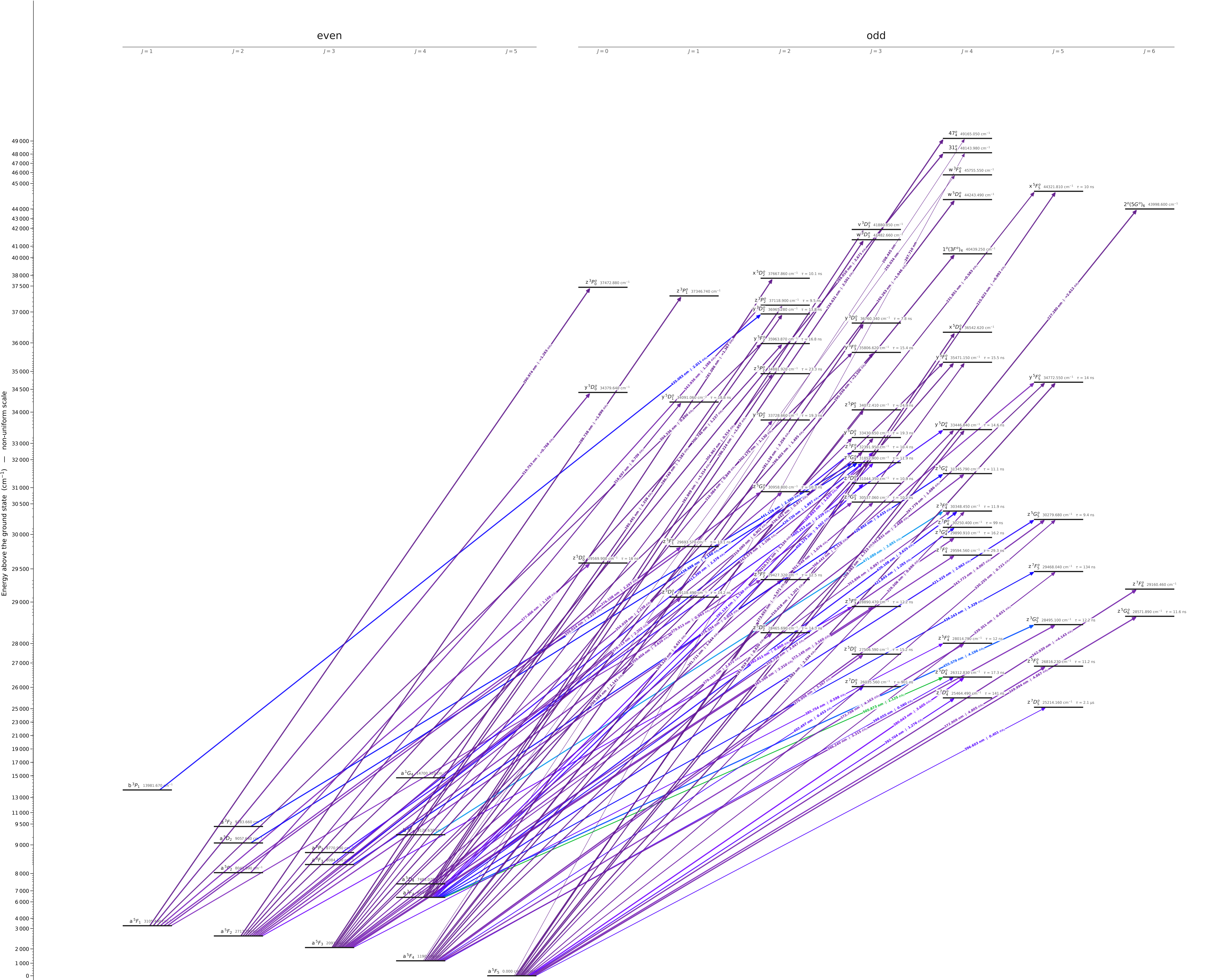
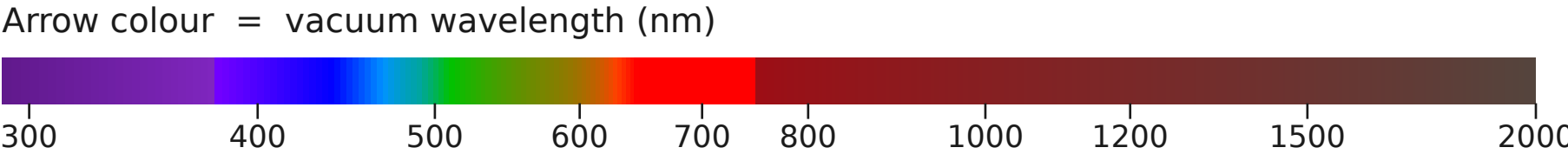


Ruthenium

Ru (Ru I). Strongest classified transitions below 2 μm (NIST): 70 levels, 90 lines.

Ru + ionisation limit 59366.40 cm⁻¹ = 7.36050 eV (168.445 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element <[JferJ']>| in ea;
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model or semi-empirical calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	<[JferJ']> (eaø)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
a ⁵ F ₅ - y ⁵ F ₅ ^o	287.5832	287.4988	1042.4548	2.334	4.220e+07	11.37	59.1	first step of the two-colour two-photon resonance-ionizat...
a ⁵ F ₃ - y ⁵ F ₃ ^o	296.6032	296.5166	1010.7527	1.495	2.480e+07	10.33	38.2	high-resolution crossed-beam laser spectroscopy line
a ⁵ F ₄ - y ⁵ F ₄ ^o	297.7794	297.6925	1006.7603	1.086	8.220e+06	11.37	11.5	high-resolution crossed-beam laser spectroscopy line
a ⁵ F ₅ - y ⁵ D ₅ ^o	298.9819	298.8947	1002.7110	1.723	2.500e+07	10.9	36.5	high-resolution crossed-beam laser spectroscopy line
a ⁵ F ₃ - y ⁵ F ₃ ^o	299.5841	299.4967	1000.6955	1.141	1.090e+07	10.27	16.9	high-resolution crossed-beam laser spectroscopy line
a ⁵ F ₂ - z ⁵ F ₂ ^o	300.7462	300.6586	996.8288	1.237	2.280e+07	9.474	38.3	high-resolution crossed-beam laser spectroscopy line
a ⁵ F ₂ - y ⁵ F ₂ ^o	302.1752	302.0873	992.1146	1.130	1.340e+07	10.33	20.6	high-resolution crossed-beam laser spectroscopy line
a ⁵ F ₄ - z ⁵ F ₄ ^o	304.1199	304.0314	985.7707	0.631	4.100e+06	6.392	10.2	high-resolution crossed-beam laser spectroscopy line
a ⁵ F ₁ - y ⁵ F ₁ ^o	304.3364	304.2478	985.0694	0.890	1.140e+07	9.474	19.1	high-resolution crossed-beam laser spectroscopy line
a ⁵ F ₃ - z ⁵ F ₃ ^o	304.9675	304.8788	983.0309	0.514	3.770e+06	6.831	8.78	high-resolution crossed-beam laser spectroscopy line
a ⁵ F ₄ - y ⁵ D ₄ ^o	310.0179	309.9280	967.0166	1.201	1.090e+07	10.9	15.9	high-resolution crossed-beam laser spectroscopy line
a ⁵ F ₃ - y ⁵ D ₃ ^o	316.0844	315.9929	948.4570	0.849	9.240e+06	8.246	17.8	high-resolution crossed-beam laser spectroscopy line
a ⁵ F ₂ - y ⁵ D ₂ ^o	318.6965	318.6043	940.6834	0.709	1.050e+07	8.466	19.7	high-resolution crossed-beam laser spectroscopy line
a ⁵ F ₃ - y ⁵ D ₁ ^o	319.0901	318.9979	939.5229	0.903	7.260e+06	8.246	14	high-resolution crossed-beam laser spectroscopy line
a ⁵ F ₁ - y ⁵ D ₀ ^o	319.7529	319.6605	937.5754	≈0.768	3.660e+07	-	-	high-resolution crossed-beam laser spectroscopy line
a ⁵ F ₅ - z ⁵ F ₅ ^o	329.5061	329.4112	909.8236	0.696	3.050e+06	13.37	3.63	high-resolution crossed-beam laser spectroscopy line
a ⁵ F ₅ - z ⁵ G ₅ ^o	330.2545	330.1594	907.7620	0.721	2.660e+06	16.93	2.5	high-resolution crossed-beam laser spectroscopy line
a ⁵ F ₄ - z ⁵ G ₄ ^o	343.7721	343.6736	872.0675	4.007	7.280e+07	16.93	68.4	high-resolution crossed-beam laser spectroscopy line
a ⁵ F ₅ - z ⁵ F ₅ ^o	349.9943	349.8942	856.5637	4.867	8.610e+07	13.72	99.9	reference line for collinear laser spectroscopy of Ru cha...
a ⁵ F ₁ - z ⁵ G ₁ ^o	359.0238	358.9213	835.0212	3.225	9.110e+07	15.45	93.8	high-resolution crossed-beam laser spectroscopy line
a ⁵ F ₂ - z ⁵ F ₂ ^o	359.4043	359.3017	834.1371	3.620	8.170e+07	15.6	83.3	high-resolution crossed-beam laser spectroscopy line
a ⁵ F ₃ - z ⁵ G ₃ ^o	359.7204	359.6178	833.4041	2.947	4.200e+07	9.824	68	high-resolution crossed-beam laser spectroscopy line

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

¹⁰¹Ru hyperfine structure (I = 5/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
a ⁵ F ₅	-229.288(3)	158.93(6)	F=5/2: +3508.19 F=7/2: +2657.47 F=9/2: +1582.77 F=11/2: +298.37 F=13/2: -1178.23 F=15/2: -2826.37
a ⁵ F ₄	-183.474(4)	101.80(5)	F=3/2: +2343.42 F=5/2: +1854.74 F=7/2: +1182.04 F=9/2: +336.77 F=11/2: -666.34 F=13/2: -1809.29
a ⁵ F ₃	-151.350(4)	59.32(5)	F=1/2: +1549.10 F=3/2: +1306.05 F=5/2: +906.91 F=7/2: +360.58 F=9/2: -320.50 F=11/2: -1120.30
a ⁵ F ₂	-92.4974(27)	31.869(23)	F=1/2: +669.79 F=3/2: +516.70 F=5/2: +269.52 F=7/2: -59.79 F=9/2: -454.52
a ⁵ F ₁	119.63(20)	39.0(3)	F=3/2: -391.43 F=5/2: -150.81 F=7/2: +308.82
z ⁵ G ₅ ^o	-97.10(10)	350(4)	F=7/2: +1838.30 F=9/2: +1294.08 F=11/2: +672.63 F=13/2: +0.17 F=15/2: -692.33 F=17/2: -1369.10
z ⁵ G ₄ ^o	-128.8(5)	125(9)	F=3/2: +1671.48 F=5/2: +1312.59 F=7/2: +824.23 F=9/2: +220.49 F=11/2: -480.54 F=13/2: -1256.70
z ⁵ G ₃ ^o	-74.50(20)	237(6)	F=5/2: +1220.03 F=7/2: +887.51 F=9/2: +488.38 F=11/2: +43.92 F=13/2: -419.82 F=15/2: -872.10
z ⁵ F ₅ ^o	-87.3(3)	16(9)	F=3/2: +1099.25 F=5/2: +876.20 F=7/2: +565.76 F=9/2: +169.77 F=11/2: -309.42 F=13/2: -868.92
z ⁵ G ₂ ^o	-123.4(3)	178.9(27)	F=1/2: +1341.34 F=3/2: +1107.94 F=5/2: +736.82 F=7/2: +254.83 F=9/2: -300.47 F=11/2: -880.77
z ⁵ G ₁ ^o	-249.1(5)	162.8(28)	F=1/2: +1857.66 F=3/2: +1410.75 F=5/2: +706.60 F=7/2: -193.74 F=9/2: -1204.80
y ⁵ D ₅ ^o	-149.30(10)	-149.7(28)	F=1/2: +1403.18 F=3/2: +1219.65 F=5/2: +898.79 F=7/2: +418.16 F=9/2: -253.69 F=11/2: -1157.17
y ⁵ D ₄ ^o	-175.20(20)	-11(4)	F=3/2: +2184.75 F=5/2: +1749.90 F=7/2: +1139.91 F=9/2: +353.57 F=11/2: -610.66 F=13/2: -1754.67
y ⁵ D ₃ ^o	-117.80(10)	-47.0(6)	F=1/2: +791.70 F=3/2: +636.15 F=5/2: +365.15 F=7/2: -38.92 F=9/2: -600.75
z ⁵ P ₅ ^o	-229.5(3)	18(7)	F=1/2: +2305.56 F=3/2: +1956.56 F=5/2: +1376.65 F=7/2: +568.47 F=9/2: -464.28 F=11/2: -1716.85
y ⁵ D ₂ ^o	-109.70(10)	39.6(9)	F=3/2: +411.67 F=5/2: +78.02 F=7/2: -264.35
y ⁵ D ₁ ^o	-221.4(4)	491(17)	F=5/2: +3533.77 F=7/2: +2609.93 F=9/2: +1481.06 F=11/2: +191.35 F=13/2: -1205.20 F=15/2: -2644.75
z ⁵ P ₄ ^o	-162.3(4)	-50(6)	F=1/2: +1100.75 F=3/2: +880.02 F=5/2: +499.52 F=7/2: -59.69 F=9/2: -824.12
y ⁵ F ₅ ^o	-147.40(20)	271(4)	F=3/2: +1975.48 F=5/2: +1527.19 F=7/2: +930.05 F=9/2: +214.53 F=11/2: -580.22 F=13/2: -1406.30
y ⁵ F ₄ ^o	-123.9(4)	136(8)	F=1/2: +1320.72 F=3/2: +1098.10 F=5/2: +740.68 F=7/2: +268.89 F=9/2: -288.66 F=11/2: -895.20
y ⁵ F ₃ ^o	-82.4(4)	96(10)	F=1/2: +643.72 F=3/2: +477.10 F=5/2: +223.30 F=7/2: -81.83 F=9/2: -388.10

Measured A, B constants; sources in the data file.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
a ⁵ F ₅ - y ⁵ F ₅ ^o	296.603	102-104	79.5(16)
a ⁵ F ₃ - y ⁵ F ₃ ^o	296.603	101-104	103(4)
a ⁵ F ₃ - y ⁵ F ₃ ^o	296.603	100-104	168.2(28)
a ⁵ F ₃ - y ⁵ F ₃ ^o	296.603	99-104	193(5)
a ⁵ F ₃ - y ⁵ F ₃ ^o	296.603	98-104	264.6(25)
a ⁵ F ₃ - y ⁵ F ₃ ^o	296.603	96-104	341(4)
a ⁵ F ₄ - y ⁵ F ₄ ^o	297.779	102-104	-11(8)
a ⁵ F ₄ - y ⁵ F ₄ ^o	297.779	101-104	-37(8)
a ⁵ F ₄ - y ⁵ F ₄ ^o	297.779	100-104	-5(15)
a ⁵ F ₄ - y ⁵ F ₄ ^o	297.779	99-104	7(8)
a ⁵ F ₄ - y ⁵ F ₄ ^o	297.779	98-104	44(16)
a ⁵ F ₄ - y ⁵ F ₄ ^o	297.779	96-104	27(27)
a ⁵ F ₅ - y ⁵ D ₅ ^o	298.982	102-104	137.6(8)
a ⁵ F ₅ - y ⁵ D ₅ ^o	298.982	101-104	213.6(15)
a ⁵ F ₅ - y ⁵ D ₅ ^o	298.982	100-104	277.2(14)
a ⁵ F ₅ - y ⁵ D ₅ ^o	298.982	99-104	356.0(17)
a ⁵ F ₅ - y ⁵ D ₅ ^o	298.982	98-104	426.4(14)
a ⁵ F ₅ - y ⁵ D ₅ ^o	298.982	96-104	579.6(19)
a ⁵ F ₃ - y ⁵ F ₃ ^o	299.584	102-104	46(5)
a ⁵ F ₃ - y ⁵ F ₃ ^o	299.584	101-104	44(3)
a ⁵ F ₃ - y ⁵ F ₃ ^o	299.584	100-104	110(4)
a ⁵ F ₃ - y ⁵ F ₃ ^o	299.584	99-104	111(5)

Shift = ν(first isotope) - ν(second isotope).